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The effects of fatigue and residual strain on the mechanical behavior of poly(ethylene terephthalate) unreinforced and nanocomposite fibers

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Poly(ethylene terephthalate) (PET) control fibers (nominal diameter $\sim$24 ± 3 μm) and PET fibers with embedded vapor-grown carbon nanofibers (PET-VGCNF) (nominal diameter $\sim$25 ± 2 μm) were exposed to cyclic loading and monotonic tensile tests. The control fibers were processed through a typical melt-blending technique and the PET-VGCNF samples were processed with approximately 5 wt.% carbon nanofibers present in the sample. Under uniaxial fatigue conditions, the fibers were subjected to a maximum stress that was approximately 60% of the fracture stress of the sample at an elongation rate of 10 mm/min in uniaxial tension. The fibers were subjected to a frequency of 5 Hz. Subsequent to non-fracture fatigue conditions, the fibers were tested under uniaxial stress conditions for observation of the change in mechanical properties to assess the effects of fatigue loading. The elastic modulus, hardening modulus, fracture strength, work done, and yield strain of both PET control and PET-VGCNF samples in uniaxial tension subsequent to fatigue were shown to be dependent on the residual fatigue strains. Relative mechanical properties were used to quantify the difference in PET and PET-VGCNF samples as a function of residual strain. In most cases, the results indicated a strengthening mechanism (strain hardening effect) in the low residual strain limit for fatigued PET samples and not for fatigued PET-VGCNF samples. In comparison with the unreinforced PET sample, the PET-VGCNF fibers showed greater degradation of mechanical properties as a function of residual strain due to fatigue when cycled at 60% of the fracture stress. The effects of the fatigue process on the change in mechanical properties have been quantified and supported through existing qualitative, quantitative, and scanning electron microscopy (SEM) techniques.

1. Introduction

Poly(ethylene terephthalate) (PET) fibers have been employed as reinforcing agents in many applications. They are especially known for their toughness and high strength-to-weight ratio. As engineers seek to develop stronger materials for advanced applications, the inclusion of reinforcing agents presents itself as a viable option for increased strength. The field of nanocomposites is particularly attractive for engineers and designers in the field nanotechnology and mechanics of materials. Nano-sized reinforcing agents such as single-walled carbon nanotubes (SWNTs), double-walled nanotubes (DWNTs), multi-walled nanotubes (MWNTs), and vapor-grown carbon nanofibers (VGCNFs) are all candidates for increasing the mechanical properties of various polymer matrices. In this investigation, the mechanical properties of nanocomposite fibers were investigated from a residual standpoint: how the fatigue process engendered microstructural changes in the fibers and altered their mechanical properties. Limited research exists on the deformation mechanisms of fatigue in nanocomposite materials, motivating this study.

In a recent investigation, Ma et al. [1] utilized various compounding methods (ball milling, high shear mixing, and extrusion) and a traditional fiber spinning method to develop PET-VGCNF composite fibers. In the case of the heat-treated nanocomposite fibers under uniaxial tensile loading at a fixed strain rate, the elastic modulus was shown to be slightly higher than the PET control samples, and the fracture strength was slightly lower than the PET control sample. In the current investigation, the same fibers that were tested in [1] were used to determine their mechanical resistance to cyclic loading under various loading conditions.

The ultimate goal of the current research is to describe the evolution of change that occurs during uniaxial cyclic loading for PET nanocomposite fibers and their unreinforced counterparts from the standpoint of changes in mechanical properties. Another goal is to ascertain the similarities/dissimilarities in the changes of mechanical properties of PET nanocomposite and unreinforced filaments with respect to residual strain, if any. Many studies have been conducted on nanocomposite samples under simple loading condi-

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tions to ascertain the effects of filler content on the mechanical properties. Sandler et al. [2] have performed uniaxial tensile experiments on melt-spun polyamide 12 fibers employed with various reinforcing agents to include arc-grown nanotubes (AGNT), aligned catalytically grown nanotubes (aCGNT), entangled catalytically grown nanotubes (eCGNT) and catalytically grown nanofibers (CNF). In all cases, the modulus and yield stress of the nanocomposites were shown to be higher than the unreinforced polyamide 12 fiber, and the values were shown to be linearly correlated with the filler content (increases in modulus and yield strength with increased filler content). The eCGNT reinforced polyamide 12 composites showed the most significant improvements in modulus (1.6 GPa) and yield strength (45 MPa) at a filler content of 10% in comparison with a modulus of 0.8 GPa and yield strength of 21 MPa for the unreinforced polyamide 12. Breton et al. [3] have also noticed significant increases in modulus with decreases in the ultimate strain and fracture strength for epoxy/MWNT composites. This clearly indicates that filling the epoxy with MWNTs leads to stiffer and more brittle materials. Other evidence has been provided by Wuite and Adali [4], Chen and Tao [5], and Kim et al. [6] that indicates stiffening of the polymer matrix due to the inclusion of nano-sized reinforcing agents under simple loading conditions.

In regards to fatigue, some interesting observations have been made regarding the physical degradation of unreinforced PET fiber samples [7–9]. In addition, Cho et al. [10] studied the fatigue behavior of unreinforced PET fibers under various processing conditions with the same crystal structure at 10^4–10^6 cycles. Thermoluminescence (TL) glow experiments under various fatigue proved that the strain hardening occurred at early stages of the fatigue process and defect sites were formed at the later stages in the fatigue process. They showed that the strain hardening effect altered the stress–strain curve of PET in the early stages of fatigue; however, its effect attenuated after a certain point in the cyclic experiments. They also showed that the viscosity molecular weights were reduced with the increase of the number of fatigue cycles. This effect was attributed to chain scission of the PET molecules during cycling. Other efforts to illuminate the effects of fatigue on the accumulation of damage in PET fibers have been investigated [11], in which destructive tests were performed. In their experiments, the ultimate failure of PET fibers after 4.22 million cycles was due to the presence of an inherent flaw attributed to antimony trioxide (Sb_2O_3) catalyst particles. Liang et al. [12] investigated the effects of chain rigidity on the non-linear viscoelastic behavior of several polymeric fibers, including PET. They concluded that the non-linear viscoelasticity was strongly governed by the rigidity of the chain, with semirigid structures such as PET exhibiting a NVP (non-linear viscoelastic parameter) between flexible polymeric fibers (Nylon 6 and PVA) and rigid polymers (Vectran and Kevlar). In essence, they showed that NVP increased with increasing chain rigidity. Liang et al. [12] also concluded that irreversible structural changes occurred faster in polymeric fibers with higher NVP values, thus indicating shorter fatigue lifetimes with increasing molecular chain rigidity. Le Clerc et al. [13] have investigated the response of mechanical properties to changes in temperature for unreinforced PET fibers and assemblies under various loading conditions. Specifically, they determined that for fiber assemblies initially at room temperature (~20 °C), a temperature rise was observed during fatigue experiments conducted at 50 Hz. The temperature rise was shown to be an increasing function of the maximum stress/load amplitude during fatigue loading and was also dependent on the median stress value (for the same load amplitude, higher maximum temperatures were observed for lower mean loads). Lechat et al. [7] conducted fatigue experiments on polyethylene naphthalate and PET fibers at 50 Hz and compared those with creep experiments to demonstrate that static creep lifetimes tested at 70% of the fracture strength were much higher than cyclic lifetimes tested under stress-controlled conditions.

In terms of polymeric composites, Mallick and Zhou [14] have examined the effect of the mean stress on the fatigue performance of nylon 66 composite fibers reinforced with E-glass fibers. By performing stress-controlled experiments at a maximum stress ranging from 55% to 80% of the tensile strength, an ensuing creep strain was observed for the experiments at a frequency of 1 Hz for various R ($\sigma_{min}/\sigma_{max}$) values ranging from 0.1 to 0.8. The fatigue creep was shown to be closely correlated with creep engendered from static creep tests. They performed experiments that corroborated the similarity between the creep rupture time curve (static) and the *S–N* curve (dynamic). Bellemare et al. [15] and Juwono and Edward [16] have investigated the mechanical behavior/fatigue properties of clay nanocomposites. Due to the inclusion of clay nanoparticles, noticeable gains were observed in the mechanical properties under fatigue loading conditions to ultimate failure. One noticeable observation from the literature results on fatigue is that the load ratios are typically defined in terms of the maximum (tensile) strength of the material. However, as will be shown in this research, defining maximum fatigue stresses in terms of the maximum (tensile) strength could possibly lead to disparate results when comparing unreinforced and reinforced (composite) materials.

The literature clearly identifies the ultimate failure of composites and unreinforced materials after some prescribed number of cycles; however, few results exist to evince how the mechanical properties change under various loading conditions. The research in this paper will identify how the residual strain that remains as a result of load-controlled fatigue conditions for different load ratios is a key parameter in determining the evolution of change in the mechanical properties.

2. Experimental methods

2.1. Sample preparation

The PET-VGCNF specimens were prepared by the Kumar Research Group at Georgia Institute of Technology. The experimental procedure employed in the production of these nanocomposite fibers as well as basic mechanical properties are given in [1]. Single PET-VGCNF filaments were cut to obtain a gage length of 1″ (25.4 mm). The single fibers were bonded to a manila (paper) rectangular gasket (0.1 mm thickness) manufactured by the Miami Valley Gasket Co. (Dayton, OH, USA) via Scotch® Super Strength All Purpose adhesive (drying adhesive). The adhesive was allowed to cure for 24 h for complete bonding. The diameter of the PET control filaments was 24 ± 3 μm and the diameter of the PET-VGCNF filaments was 25 ± 2 μm.

2.2. Fatigue and uniaxial tensile tests

The BOSE® ELectroForce® (ELF®) 3200 tensile and fatigue testing machine (Enduratec) was used to conduct the mechanical experiments in uniaxial cyclic loading and uniaxial tension. The Enduratec has a maximum load of 225 N and a maximum frequency of 400 Hz. A set of low mass grips, Model GRP-TC-DMA450N from BOSE ElectroForce (Eden Prairie, MN, USA), were used. The load cell had a maximum load rating of 2.5 N (250 g) and resolution of ±10 mg. The ELF 3200 measures displacements via a Capacitec 100 μm displacement transducer (Model HPC-40/4101) used as a feedback for the control loop. The resolution of this displacement transducer was ±50 μm full-scale. All experiments were conducted at room temperature, with a typical humidity of 50%.

The fatigue experiments were conducted between a minimum and maximum load in tension for a prescribed number of cycles using a set frequency. All fatigue tests were conducted at 5 Hz at load ratios of $R = 0$ and $R = 0.333$ for the nanocomposite and control samples. Here, the load ratio R is defined as the ratio of the minimum and the maximum stress (load) ($R = \sigma_{min}/\sigma_{max}$) during fatigue. The maximum fatigue load at which the samples were subjected to was 58% of the fracture stress for PET-VGCNF samples and 57% of the estimated fracture stress of PET control samples (approximately 60% of fracture stress in each case). At the conclusion of the fatigue phase, a displacement (strain) controlled condition was imposed to unload the specimen at an elongation rate of −10 mm/min. The fiber specimens were monotonically loaded to zero applied stress and held at that state for 10 s. A residual strain remained in the samples as a result of the constant load fatigue process. At the conclusion of the dwell phase, the filaments were loaded to failure (or a maximum load within the stroke limitations of the Enduratec) at an elongation rate of 10 mm/min. The residual strain was determined by considering the displacement reading on the oscilloscope at the initiation of the test and at the conclusion of the dwell phase. The PET and PET-VGCNF filament can be idealized as a viscoelastic solid that was loaded in constant stress fatigue and obtained an initial elastic strain e_1, a delayed elastic strain e_2, and a viscous strain, e_3, as depicted in Fig. 1. When unloaded, the residual strain, e_3, that remained is due to the viscous nature of the polymeric solid. The residual strain values for the PET control samples was in the range $0 \leqslant \varepsilon_R \leqslant 0.23$ and for the PET-VGCNF samples the residual strain range was $0 \leqslant \varepsilon_R \leqslant 0.16$.

Based on the terminology and depiction in Fig. 1, the residual strain was measured by subtracting the oscilloscope displacement value at the initiation of the test from the displacement at the conclusion of the constant-stress amplitude fatigue phase, once the specimen was unloaded to zero stress and allowed to dwell for a short time.

Approximately 150 fibers were tested in this investigation. Approximately 10% of the fibers broke at the grip interface at the conclusion of the uniaxial tensile loading phase, which indicated premature failure due to a stress concentration near the grip interface. For this reason, these experimental results were omitted from the results in this study. All fibers were subjected to the same frequency during cyclic loading (5 Hz), elongation rate during uniaxial tension (10 mm/min), and processing conditions during synthesis [1] of the nanocomposite fibers. A detailed description of the preparation of the samples for mechanical testing is provided in [18].

The dynamic stress–strain response of a viscoelastic polymer sample under fatigue conditions is given as:

$$\begin{aligned} \varepsilon &= (\Delta\varepsilon)\sin\omega t \\ \sigma &= (\Delta\sigma)\sin(\omega t + \delta) \end{aligned} \tag{2.1}$$

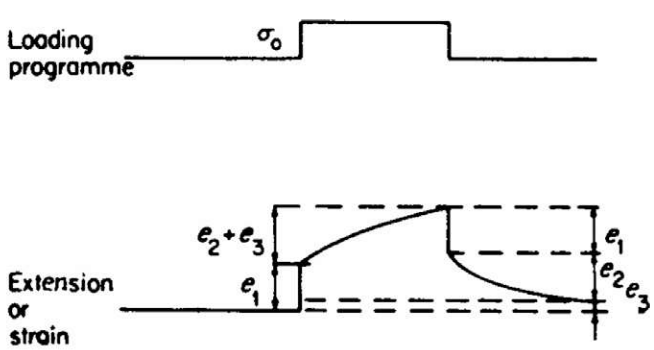

Fig. 1. Idealized deformation process in a linear viscoelastic solid depicting the residual strain that remains as a result of a constant load experiment [17].

In (2.1), ε is the instantaneous strain, $\Delta\varepsilon$ is the strain amplitude, ω is the angular frequency, σ is the instantaneous stress, $\Delta\sigma$ is the stress amplitude, and δ is the phase lag component of the stress and strain. In ordinary polymeric materials, typically the strain lags behind the stress response due to the viscous component of the polymer sample. One can envision that this phase lag δ of the strain component is the direct result of the imaginary viscous dashpot inhibiting the strain function from being directly in phase with the applied sinusoidal stress. One caveat that must be noted is that Eq. (2.1) is only valid under equilibrium conditions, once the stress and strain have both attained fixed amplitude values for the duration of the fatigue test. At the initiation of the fatigue test, Eq. (2.1) was not valid because the strain response of the material was changing non-linearly with time (creeping), due to the application of an imposed constant stress. This phenomenon is very similar to the creep behavior in polymers exposed to a constant stress for a long period of time (Fig. 1).

3. Results and discussion

3.1. Uniaxial tensile tests (specimens without prior fatigue)

The response of the PET and PET-VGCNF fibers without prior fatigue was non-linear elastic, strain hardening (see Fig. 6). Subsequent to the yield point, the fiber began to strain harden non-linearly, and finally reached the ultimate tensile strength or the maximum stress within the stroke limitations of the machine. The experimental material parameters at an elongation rate of 10 mm/min were determined and are shown in Tables 1 and 2.

A 95% confidence interval was provided to determine the range of values within which the mean value was likely to fall within. Although the samples are identical to the samples tested in [1], the elongation rate in the current study was approximately 40% of the elongation rate in that study. This validates the minor differences in the mechanical properties for the two studies in the case of the unfatigued samples. As shown in Tables 1 and 2, the PET nanocomposite sample (PET-VGCNF) exhibited higher mechanical properties than the PET control sample in terms of elastic modulus (E), hardening modulus (H), yield strength (σ_y), and tensile energy. The unfatigued yield strains of the PET-VGCNF and PET control samples were similar. In terms of maximum strains within the stroke limitations of the machine (~12 mm), the PET control sample exhibited a maximum strain (ε_m) equivalent to 0.45 ± 0.017.

Determination of the yield point was an important parameter for characterizing the PET and PET-VGCNF samples under uniaxial tension. As described earlier, the non-linearity of the stress–strain curve obscured the exact value of the yield point; however, the method as described in [19] was utilized to determine this value. Two intersecting lines based on the modulus value up to 1% strain and a hardening modulus in the later plastic stages (5% less than ε_m up to ε_m) were employed in the calculations for consistent determination of the yield stress based on the following set of equations:

From the elastic modulus:

$$\sigma_E = \left.\frac{d\sigma}{d\varepsilon}\right|_{\varepsilon=0}^{\varepsilon=0.01} \varepsilon + b_1 \tag{3.2}$$

From the hardening modulus:

$$\sigma_H = \left.\frac{d\sigma}{d\varepsilon}\right|_{\varepsilon=\varepsilon_{max}-0.05}^{\varepsilon=\varepsilon_{max}} \varepsilon + b_2 \tag{3.3}$$

In (3.2) and (3.3), σ_E and σ_H represent the linearized form of the elastic stress and hardening stress evaluated between the given limits and b_1 and b_2 are arbitrary intercepts. These equations were set equivalent to one another and the yield stress and strain were defined by the vertical intersection line to the stress–strain curve.

Table 1
Properties of PET-VGCNF filaments at elongation rate of 10 mm/min under uniaxial stress conditions without prior fatigue.

Fiber name	E (GPa)	E (GPa) secant	H (MPa)	σ_0 (MPa)	σ_f (MPa)	ε_0	ε_f	Energy (μJ)
PET-VGCNF 1	15.2	10.9	370	280	510	0.027	0.29	1.4E+03
PET-VGCNF 2	16.0	11.7	370	301	550	0.026	0.31	1.8E+03
PET-VGCNF 3	15.8	10.6	380	319	550	0.030	0.33	1.9E+03
PET-VGCNF 4	15.9	11.4	330	297	520	0.026	0.29	1.6E+03
PET-VGCNF 5	16.2	10.9	430	315	500	0.033	0.20	1.0E+03
Average	15.8	11.1	380	300	530	0.028	0.28	1.5E+03
Std. dev.	0.38	0.47	36	15	23	0.0030	0.050	3.5E+02
95% interval	0.47	0.58	44	19	29	0.0037	0.062	4.4E+02

The variables in this table are defined as follows: E, elastic modulus up to 1% strain; E secant, secant modulus; H, hardening modulus; σ_0, yield stress; σ_f, fracture strength; σ_m, maximum stress within stroke limitations; ε_0, yield strain; ε_m, maximum strain within stroke limitations; Energy, total work done in uniaxial tension.

Table 2
Properties of control PET filaments at elongation rate of 10 mm/min under uniaxial stress conditions without prior fatigue. (The ** values indicate samples that did not fracture within the stroke limitations of the machine. In actuality, the fracture stress of the PET control sample was estimated as 290 MPa)

Fiber name	E (GPa)	E (GPa) secant	H (MPa)	σ_0 (MPa)	σ_f or σ_m (MPa)	ε_0	ε_f or ε_m	Energy (μJ)
Control 1**	6.8	4.7	230	100	250	0.018	0.46	9.87E+02
Control 2	6.8	3.9	210	120	280	0.029	0.45	1.13E+03
Control 3	7.9	5.1	280	120	270	0.022	0.45	1.13E+03
Control 4**	7.7	5.3	220	150	280	0.028	0.46	1.23E+03
Control 5**	5.0	2.4	140	90	210	0.037	0.45	9.37E+02
Control 6	5.1	3.2	160	90	200	0.028	0.41	7.96E+02
Control 7**	6.8	4.6	200	130	270	0.029	0.45	1.17E+03
Average	6.6	4.2	210	110	250	0.027	0.45	1.1E+03
Std. dev.	1.1	1.1	46	22	33	0.0059	0.019	1.5E+02
95% interval	1.1	1.0	43	21	31	0.0055	0.017	1.4E+02

The variables in this table are defined as follows: E, elastic modulus up to 1% strain; E secant, secant modulus; H, hardening modulus; σ_0, yield stress; σ_f, fracture strength; σ_m, maximum stress within stroke limitations; ε_0, yield strain; ε_m, maximum strain within stroke limitations; Energy, total work done in uniaxial tension.

3.2. Fatigue experimentation phase

In this study, fatigue conditions were applied to the specimens, similar to the research in [18]. In essence, the fatigue tests performed in this experiment were not totally destructive as in normal fatigue tests where the material is cycled to failure [11]. Rather, the fatigue cycles in these experiments were employed to make an assessment on the residual properties of the fiber subsequent to fatigue at a maximum load of approximately 60% of the fracture stress. The overall objective was to ultimately develop a correlation on the evolution of damage and change in mechanical properties of polymeric nanocomposite and pristine fibers subsequent to fatigue. In addition, another goal was to determine if the load amplitude (load ratio) had an effect on the overall mechanical response of the fibers subsequent to fatigue. As shown in Table 2, only three PET control samples fractured within the stroke limitations of the machine (** indicates samples that did not fracture), whereas the results in Table 1 for the nanocomposite (PET-VGCNF) sample (σ_{max}) do correspond to the fracture stress. Based on the samples that fractured from Table 2 and the standard deviation band, the estimated fracture stress of the PET sample was 290 MPa. This value of the PET unreinforced fiber was 55% of the fracture strength of the PET-VGCNF fiber sample.

Figs. 3 and 4 display a 1 s interval of the load vs. time and displacement vs. time response of the PET-VGCNF sample and the control sample under sinusoidal cyclic loading conditions between 0–15 g (PET-VGCNF) and 0–8 g (control PET) after equilibrium stress and strain values were reached. These samples were subjected to uniaxial fatigue loading conditions. From the graph, the phase lag components are clearly visible. Both the neat sample and the nanocomposite specimen behaved similar to a viscoelastic solid, exhibiting small phase angle values between stress and strain for a fixed frequency of 5 Hz.

The viscoelastic behavior for both PET control and PET-VGCNF samples has been confirmed through dynamic mechanical analysis

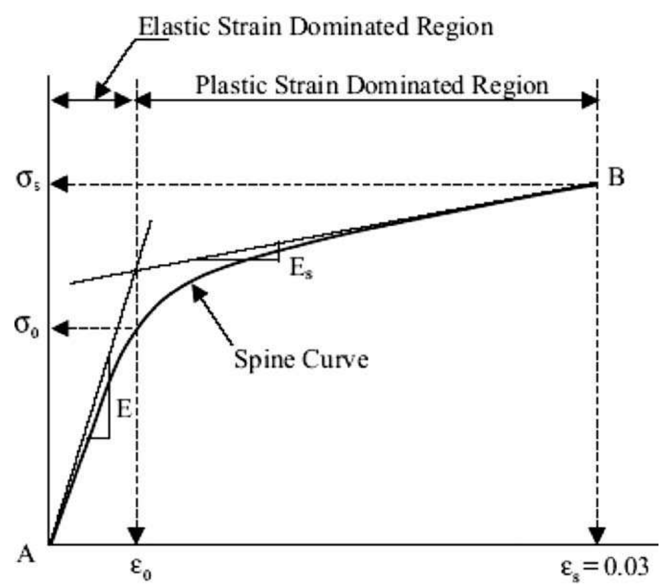

Fig. 2. Stress–strain curve for material showing elastic and plastic regions depicting a method to determine the yield stress [19].

(DMA) studies in [1], in which the authors have examined the effects of tan δ vs. temperature for a frequency of 1 Hz and observed minute differences (Fig. 5). Because it is known that tan δ is a function of frequency in the elastic range [17], the observations in Figs. 3 and 4 warrant a detailed DMA investigation at the 5 Hz frequency over a variation of temperatures for the PET and PET-VGCNF samples.

3.3. Tensile tests subsequent to load-controlled fatigue

A representation of the stress–strain response of fibers tested in uniaxial tension without prior fatigue and subsequent to fatigue is

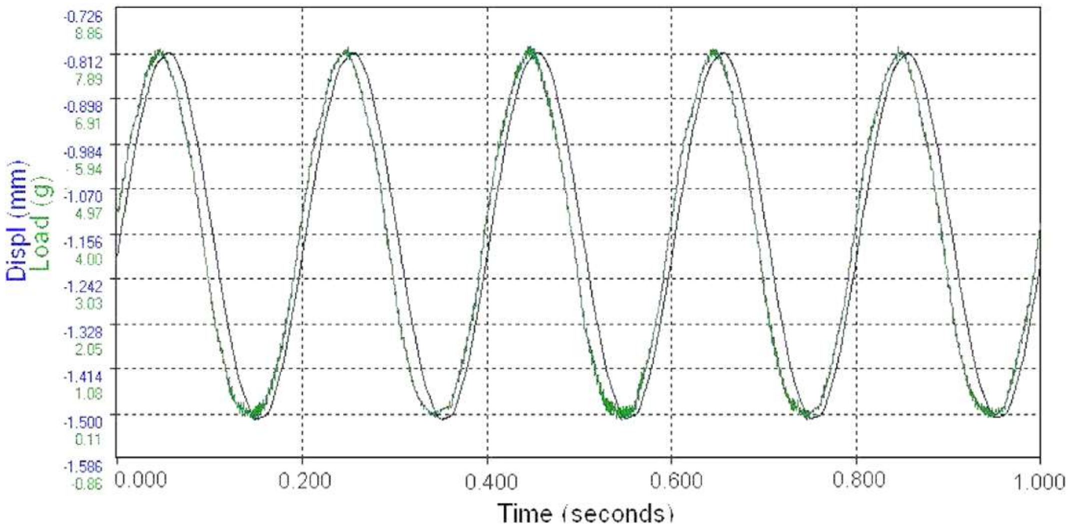

Fig. 3. Oscilloscope output of load and displacement vs. time response of PET control sample undergoing uniaxial sinusoidal loading (prescribed load values: 0–8-g stress ratio = 0 ($R = 0$)).

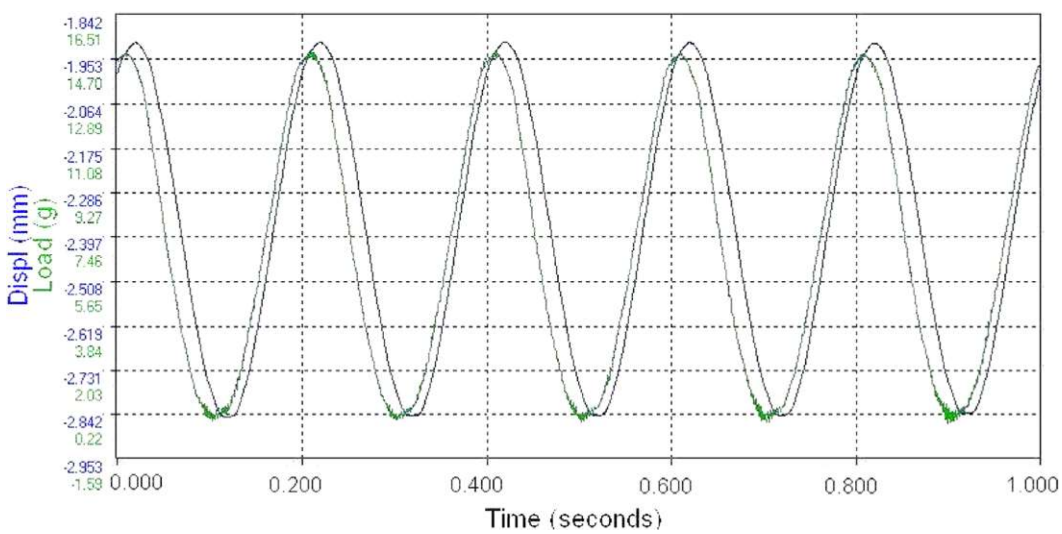

Fig. 4. Oscilloscope output of load and displacement vs. time response of PET-VGCNF sample undergoing uniaxial sinusoidal loading (prescribed load values: 0–15-g stress ratio = 0 ($R = 0$)).

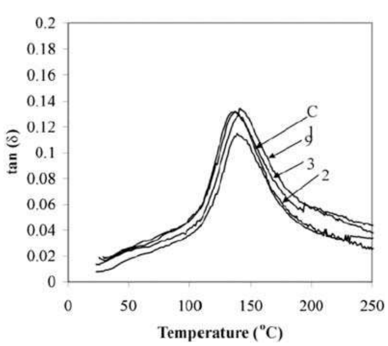

Fig. 5. Dynamic mechanical analysis properties for PET control (C) and PET-VGCNF samples (9) [1].

shown in Fig. 6. The fibers that were exposed to fatigue were shifted according to the residual strain that remained as a result of the fatigue process.

For clarity, only one stress–strain curve is shown for each corresponding loading configuration in Fig. 6. This figure is shown to elucidate the post-fatigue effects on the uniaxial stress–strain response of PET and PET-VGCNF filaments. From Fig. 6, the following observations can be made regarding the unfatigued samples vs. the fatigued samples:

(1) For PET unreinforced fibers, the stress–strain response changed from non-linear-elastic strain hardening to non-linear-elastic strain hardening with a more clearly defined yield point (piecewise linear elastic, strain hardening) and decreasing hardening modulus. The PET-VGCNF samples exhibited a slight non-linearity without prior fatigue. Similar to the PET control samples, the constitutive response of the PET-VGCNF samples changed to piecewise linear elastic, strain hardening with a more clearly defined yield point.
(2) Subsequent to fatigue, there was a generally a decrease in stress values which resembled a small "dip" in the stress–strain curve, similar to the behavior of some metals (see circled regions in Fig. 6). This "dip" has been qualitatively

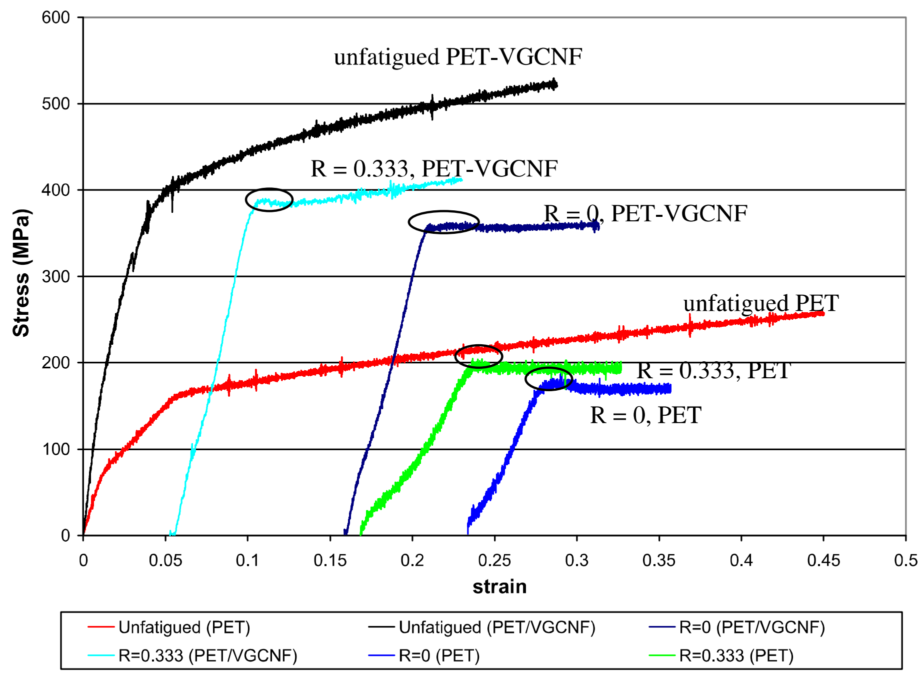

Fig. 6. Residual stress–strain response of fibers in uniaxial tension subsequent to fatigue under various loading configurations. *Note:* Stress–strain curves have been shifted according to the residual strain due to the fatigue process. The circles highlight a dip in stress values.

explained in [20] for poly(ethylene terephthalate) samples tested in uniaxial tension and was attributed to an intrinsic yield process and/or a decrease in the cross-sectional area of the specimen (necking).

(3) There was a reduction in hardening modulus for the fatigued samples vs. the unfatigued samples for PET and PET-VGCNF samples; this resembled a more perfectly plastic behavior subsequent to the static yield point, as seen in Fig. 6. In essence, the hardening modulus gradually approached a horizontal slope as the residual strain increased. This will be elaborated upon in the discussion of the change in mechanical properties.

These observations clearly show the alterations in the uniaxial response as a result of fatigue for PET and PET nanocomposite samples. Effectively, there were changes in the overall constitutive response of both samples. There were changes in the maximum stress, elastic modulus, tensile energy, hardening modulus, and the yield strain of the samples, which will be delineated in Figs. 7, 8 and Figs. 10–17. In those figures the *bold* line at the relative value = 1 indicates the absolute value point for the unfatigued sample (refer to Tables 1 and 2 for absolute values of the nanocomposite fiber and unreinforced fiber, respectively). Arrows pointing up (↑) or down (↓) depict the increase or decrease in the mechanical property value in the low residual strain limit, respectively. The relative values in Figs. 7, 8 and Figs. 10–17 were derived from the ratio of the fatigued mechanical property value to the unfatigued property value. The data have been presented in this order to indicate changes in the fracture resistance (maximum stress), elastic behavior (elastic modulus), and plastic behavior (energy absorption, hardening modulus, and yield strain) as a function of the residual strain accumulated during the fatigue process.

3.3.1. Relative tensile stress subsequent to fatigue

Fig. 7 displays relative maximum stress values (within the stroke limitations of the fatigue machine) from the PET control and PET-VGCNF samples as a function of residual strain from the fatigue process. For both the PET control and PET-VGCNF samples, the results show a decreasing trend of fracture strength vs. accumulated strain. The results from Fig. 7 show that the fatigue process conducted under $R = 0$ conditions engendered irreparable damage to the sample with the accumulation of strain for PET-VGCNF samples. These results indicate that an accumulation of flaw sites may have been generated in the PET-VGCNF specimens as a result of the fatigue process. Interestingly, the results for the PET control samples in Fig. 7 indicate a slight increase in the maximum stress for low residual strain values (less than approximately 5%). In fact, for the PET control samples in the low residual strain limit for the $R = 0$ condition, five samples achieved a higher maximum stress with the retention of fatigue strain, possibly indicating a strengthening mechanism from the fatigue process. After approximately 5% residual strain, the maximum stress values for the PET control samples showed a decaying behavior. From a comparative standpoint, the results in Fig. 7 indicate that although the nanocomposite samples possessed an overall higher maximum stress value in the unfatigued state, the PET control sample possessed higher relative maximum stress values subsequent to fatigue for all residual strain values in this study.

The data in Fig. 8 are results of the relative maximum stress in uniaxial tension subsequent to fatigue for the condition $R = 0.333$. These results indicate that the unreinforced PET relative maximum stress was higher in almost all cases for the same residual strain with reference to the nanocomposite sample. Similar to the results for the loading condition $R = 0$, there was a spike in maximum stress values for low residual strains (up to approximately 7%) with an ensuing decay behavior. The results for the relative maximum stress for the PET-VGCNF sample were somewhat monotonic, with no sharp increases in relative maximum stress for low residual strain values. Combined, the results in Figs. 7 and 8 indicate that the inclusion of VGCNFs into the PET matrix adversely affected the maximum obtainable stress in uniaxial tension subsequent to fatigue for $R = 0$ and $R = 0.333$ conditions. One explanation for the increase in maximum obtainable stress (in the low residual strain limit) for PET control samples could be that the fatigue process

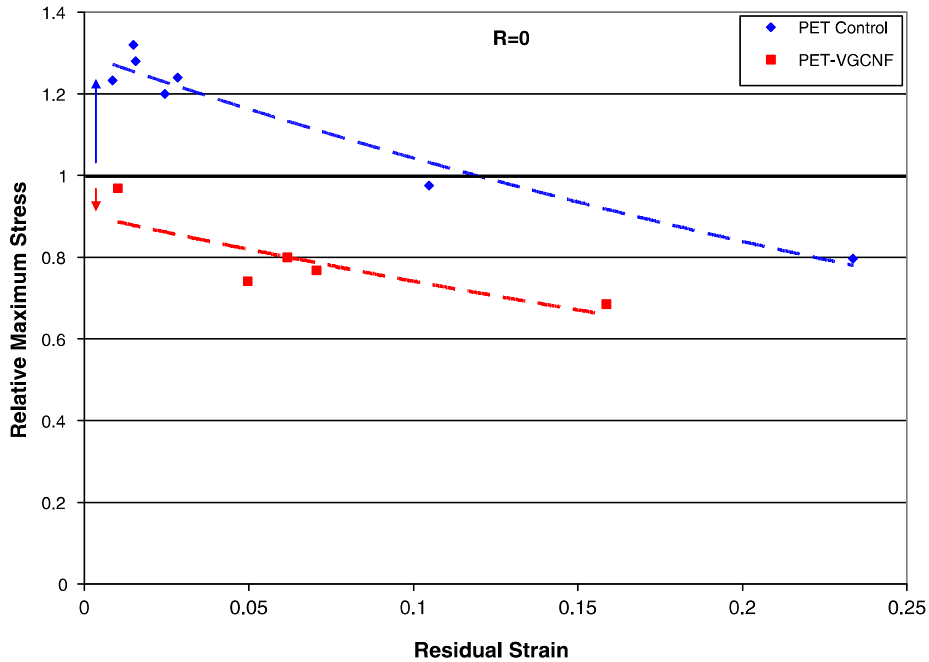

Fig. 7. Relative maximum stress of PET-VGCNF and PET control samples in uniaxial tension (subsequent to fatigue) vs. residual strain for the $R = 0$ loading condition.

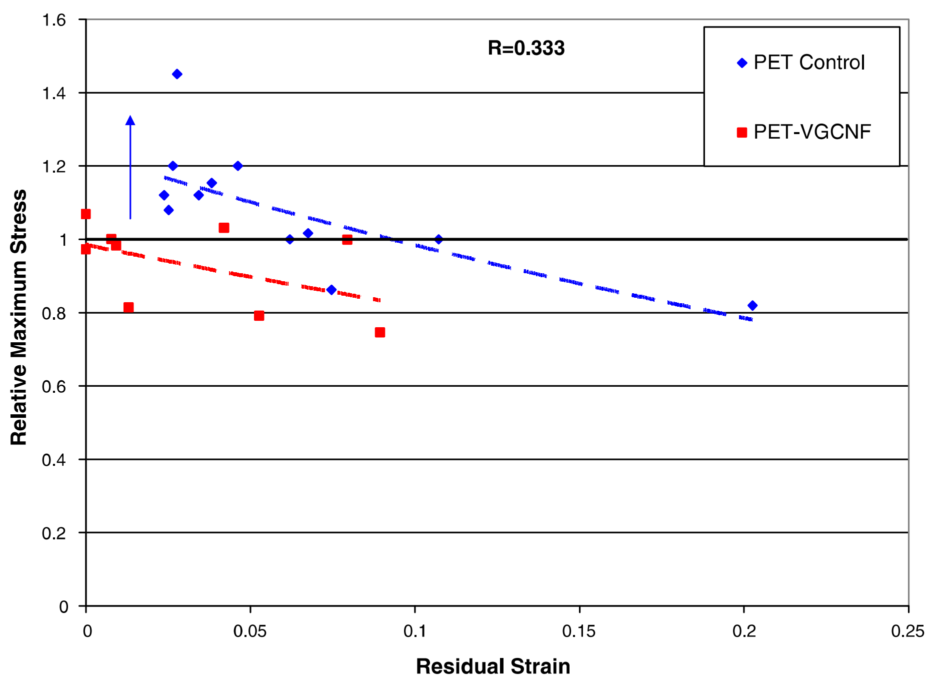

Fig. 8. Relative maximum stress of PET-VGCNF and PET control samples in uniaxial tension (subsequent to fatigue) vs. residual strain for the $R = 0.333$ condition.

engendered an alignment of the polymer chains along the main axis, causing a shift from ductile to less ductile behavior in the sample. In fact, it will be shown in the upcoming sections that for PET control samples tested subsequent to $R = 0$ fatigue loading, there was an increase in the elastic modulus and hardening modulus in the low residual strain limit, supporting the claim that the PET samples were less ductile in the low residual strain limit.

3.3.2. Elastic modulus subsequent to fatigue

There were obvious decreases in the elastic modulus (small strain limit modulus – up to 1%) for both PET and PET-VGCNF samples as a function of residual strain. A representative stress–strain curve of a PET-VGCNF sample that underwent 5000 cycles under load-controlled conditions and 3.9% accompanying residual strain is shown in Fig. 9.

This behavior is very similar to the Mullins effect [21] in which a material is loaded to a defined strain value, and then subsequently reloaded to traverse a different stress–strain curve. The effect in Fig. 9 exhibits the stress softening phenomenon as described by Mullins on the uniaxial loading of rubbers subsequent to fatigue.

Comparatively, in terms of the elastic modulus (which is an indication of the stiffness of the material), the nanocomposite

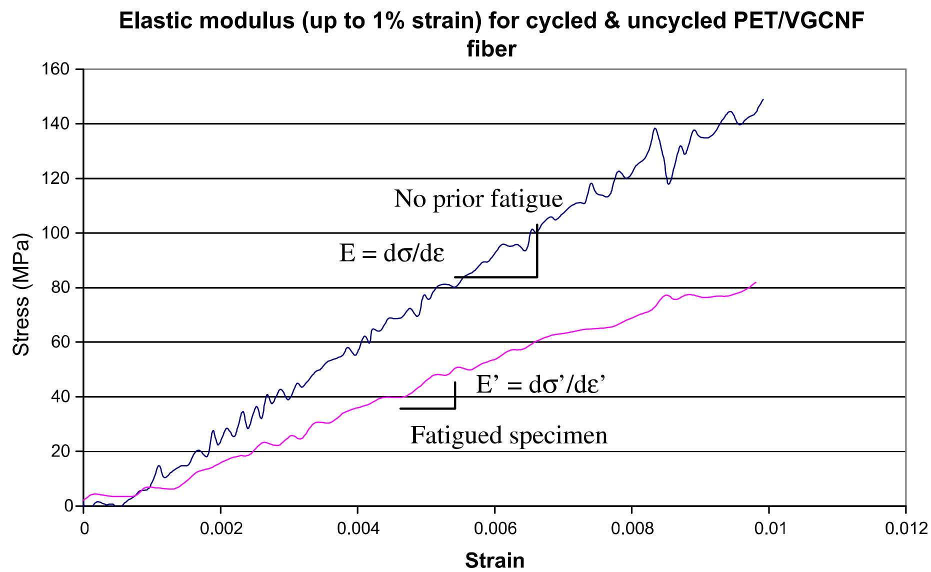

Fig. 9. Representative elastic modulus of PET-VGCNF sample before and subsequent to fatigue (up to 1% strain).

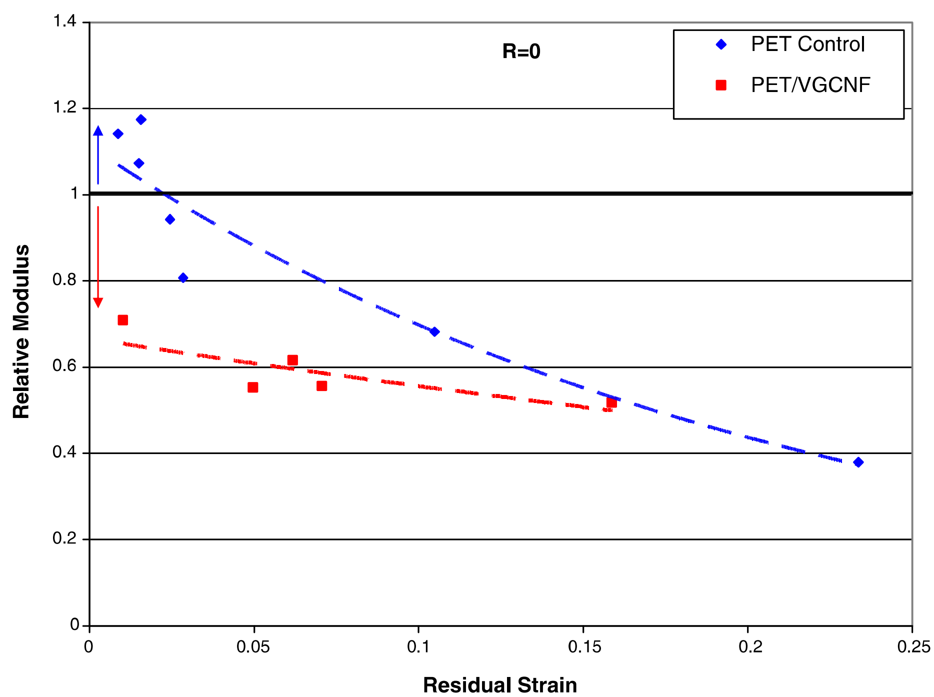

Fig. 10. Relative elastic moduli of PET control and PET-VGCNF samples (subsequent to fatigue) vs. residual strain for the $R = 0$ loading condition.

showed greater signs of modulus degradation with the retention of strain from the fatigue process. Figs. 10 and 11 depict relative moduli of the PET and PET-VGCNF samples vs. residual strain for loading conditions $R = 0$ and $R = 0.333$, respectively. In Fig. 10, the PET control sample showed a stiffening effect in the low residual strain limit, similar to the increased maximum stress behavior observed in Figs. 7 and 8. The nanocomposite samples exhibited no signs of stiffening subsequent to fatigue; rather a severe reduction in modulus was observed for small residual strain values for both $R = 0$ and $R = 0.333$ loading conditions (Figs. 10 and 11). Although the literature suggests a significant increase in modulus for nanocomposite materials [2–6] and an overall stiffer material, these results suggest that stress-controlled fatigue conditions that result in an ensuing residual strain for PET-VGNF materials can cause severe reductions to the elastic modulus. This reduction in stiffness is due to a severely degraded and defibrillated material, as will be shown in the SEM fractographs in Section 3.4.

3.3.3. Tensile energy analysis of fibers in uniaxial tension subsequent to fatigue

Another primary indicator of the damage accumulation in materials due to the fatigue process is the measurement of total energy under the load–displacement curve in uniaxial tension. This total energy represents the work done during stretching of the fiber in uniaxial tension. Measurements of the relative tensile energy were plotted vs. residual strain to determine the

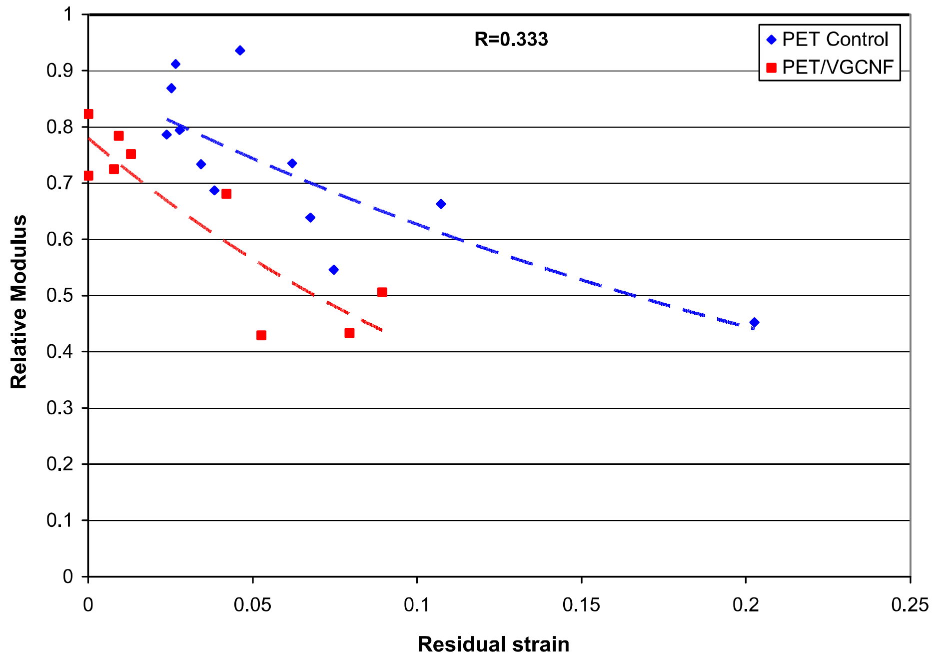

Fig. 11. Relative elastic moduli of PET control and PET-VGCNF samples (subsequent to fatigue) vs. residual strain for the $R = 0.333$ condition.

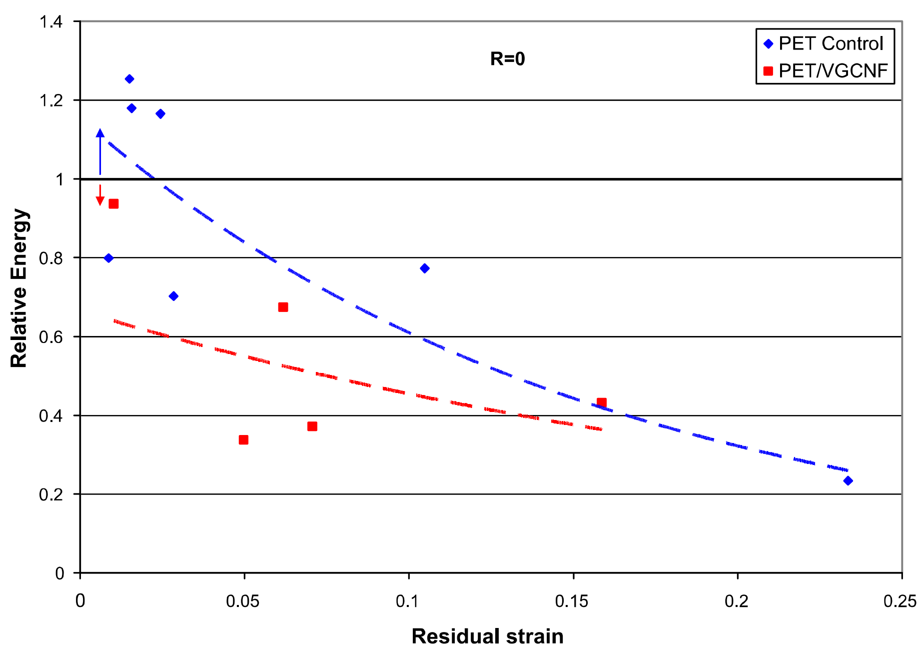

Fig. 12. Relative total energy of PET control and PET-VGCNF samples (subsequent to fatigue) vs. residual strain for the $R = 0$ loading condition.

effects of fatigue on the energy absorption capabilities of the PET control and PET-VGCNF samples. Figs. 12 and 13 display relative tensile energy results for the $R = 0$ and $R = 0.333$ condition, respectively. Although there is some scatter in the data, Figs. 12 and 13 clearly illustrate the degenerative effects from the fatigue process. These results are somewhat intuitive, as one expects less energy to be available to the sample to perform useful work after the accumulation and retention of strain from the fatigue process. After all, some of the work done was converted to hysteresis heat during the fatigue process and some was utilized for changing the underlying microstructure of the pristine sample, probably inducing polymer chain alignment along the fiber axis. Similar to the results obtained for the maximum stress and the elastic modulus, the PET control samples displayed an increase in energy absorption in the low residual strain limit for the $R = 0$ condition. In terms of tensile energy capabilities, the nanocomposite sample showed lower values with respect to the unreinforced sample for both $R = 0$ and $R = 0.333$ conditions. The results in Fig. 13 indicate that some PET-VGCNF samples displayed higher energy absorption values in the low residual strain limit; however, the trend was not consistent for similar residual strain values. There is much more scatter in the data in Fig. 13, which may have arisen from slight differences in sample structure, etc.

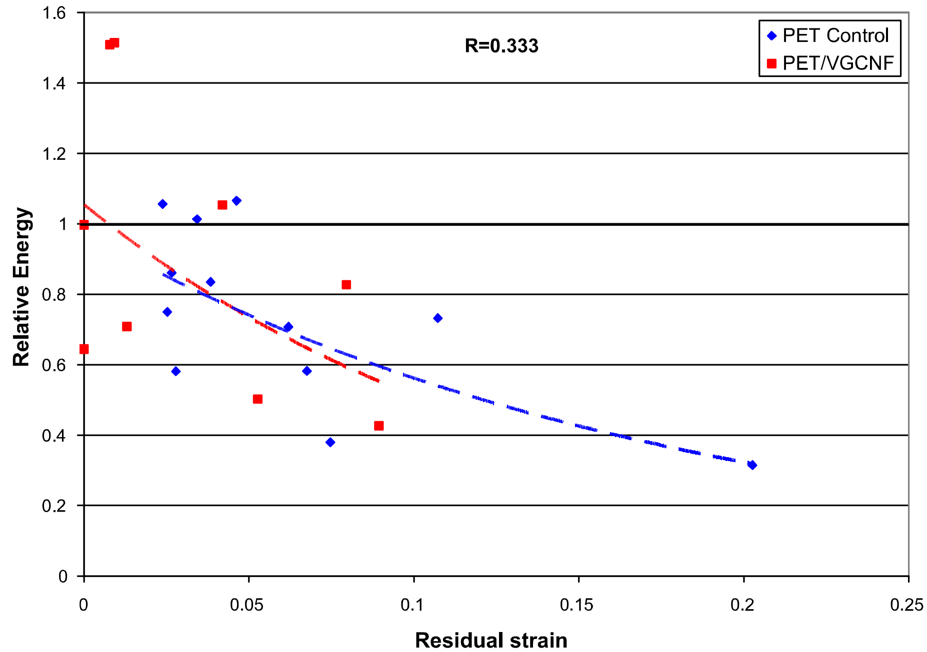

Fig. 13. Relative total energy of PET control and PET-VGCNF samples (subsequent to fatigue) vs. residual strain for the $R = 0.333$ condition.

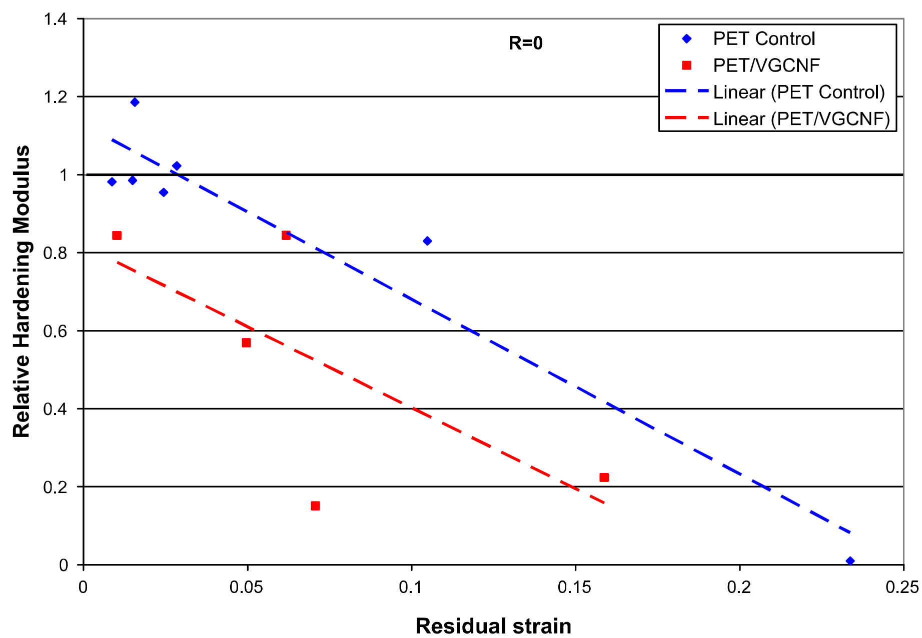

Fig. 14. Relative hardening modulus of PET control and PET-VGCNF samples (subsequent to fatigue) vs. residual strain for the $R = 0$ loading condition.

3.3.4. Hardening modulus subsequent to fatigue

The hardening behavior of samples was also studied to obtain post-yield deformation information about the samples before and subsequent to fatigue. The hardening modulus indicates the behavior of the sample in the plastic dominated region, as indicated in Fig. 2. This indicates the resistance of the material after it has yielded, where it displayed a resistance to plastic flow behavior. The results in Figs. 14 and 15 clearly indicate that the nanocomposite sample demonstrated a greater proclivity to plastic flow (less hardening) subsequent to fatigue loading at ratios of $R = 0$ and $R = 0.333$. In fact, for the PET-VGCNF sample, only one relative hardening modulus value fell above the threshold for the unfatigued sample in Figs. 14 and 15 (green line), indicating a reduction in the modulus as the result of fatigue. The PET-VGCNF hardening behavior for the $R = 0.333$ condition was fairly monotonic, with no significant increases or decreases observed as a function of residual strain. For comparison, the control PET sample showed slight increases in the hardening modulus in the low residual strain limit, further corroborating the possibility of a strengthening mechanism responsible for improved mechanical behavior in PET filaments with the retention of small strains from fatigue. This was consistent for both $R = 0$ and $R = 0.333$ loading conditions. Future studies need to be done to further quantify the mechanism responsible for the decrease in mechanical properties and overall mechanical

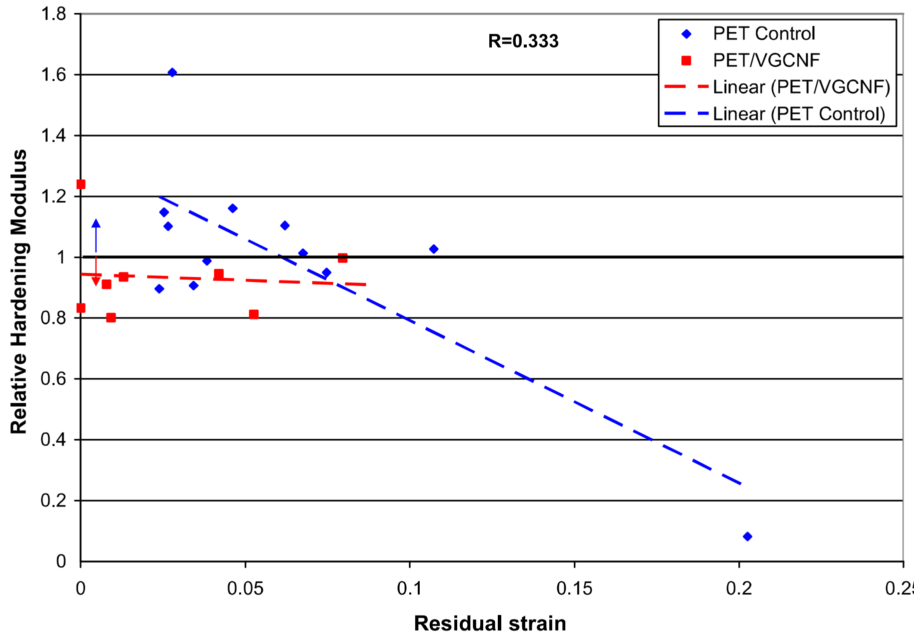

Fig. 15. Relative hardening modulus of PET control and PET-VGCNF samples (subsequent to fatigue) vs. residual strain for the $R = 0.333$ condition.

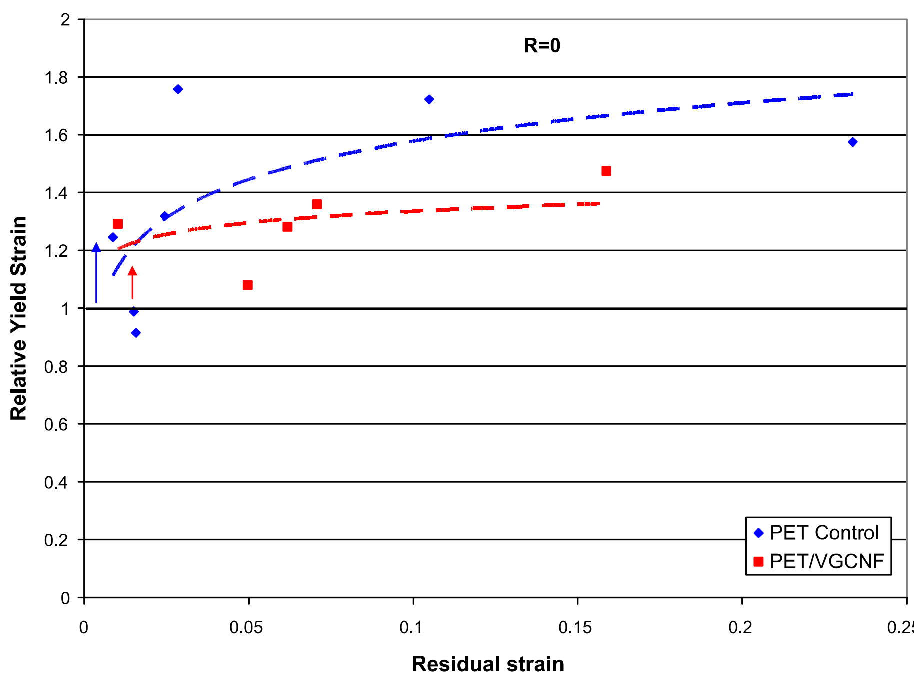

Fig. 16. Relative yield strain of PET control and PET-VGCNF samples (subsequent to fatigue) vs. residual strain for the $R = 0$ loading condition.

behavior of PET nanocomposites with the retention of strain from the fatigue process. Cho et al. [10] have already confirmed that strain hardening is a dominating effect in the early stages of cyclic extension, and defects are more dominating in the latter stages of fatigue. The results of the study from [10] are in accordance with the results in Figs. 7, 8 and Figs. 10–17 presented in the current research.

3.3.5. Yield strain subsequent to fatigue

The yield strain values of the single filaments subsequent to fatigue were also investigated to determine the effects of residual strain. Figs. 16 and 17 depict the relative yield strain as a function of residual strain for the PET control and nanocomposite sample. The results clearly show that residual strains in the material caused an increase in the yield strain of the deformed material. In essence, subsequent to fatigue, the material possessed a new length:

$$L' = L_0 + \Delta L \tag{3.4}$$

The post-fatigue engineering yield strain was defined in terms of the undeformed length as:

$$\varepsilon'_y = \frac{\delta_y}{L'} \tag{3.5}$$

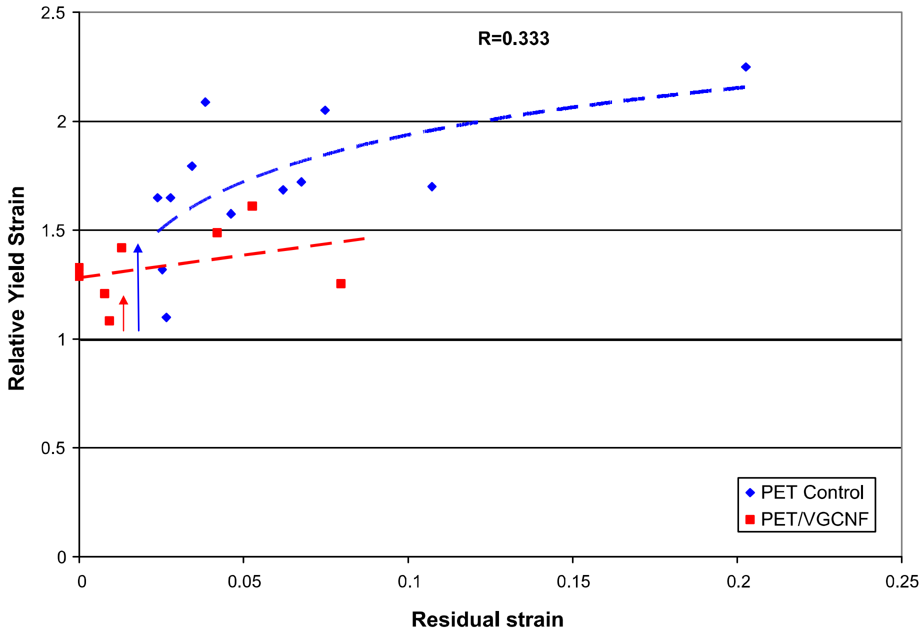

Fig. 17. Relative yield strain of PET control and PET-VGCNF samples (subsequent to fatigue) vs. residual strain for the $R = 0.333$ condition.

In (3.4) and (3.5), δ_y represents the yield displacement, L' represents the new length of the sample, L_0 represents the undeformed length of the specimen, ΔL represents the residual length that remained as a result of fatigue, and ε'_y represents the yield strain of the fatigued (deformed) sample. The results in Figs. 16 and 17 indicate that the yield strains in the fatigued PET control and PET-VGCNF (deformed) samples was inhibited as a result of residual strains from fatigue. Also, from a relative standpoint, the results seem to indicate that the relative yield strains in the PET control samples showed a greater dependence on the residual strain for both $R = 0$ and $R = 0.333$ loading conditions.

3.4. SEM fractography

Scanning electron microscopy (SEM) has been conducted to determine the effects of fatigue loading conditions on the fracture morphology of the PET unreinforced and PET-VGCNF single fibers. The results from the SEM study substantiate the quantitative results obtained from Section 3, supporting the claim that the nanocomposite fibers degraded more significantly with the accumulation and retention of residual strains than its unreinforced counterpart. Fig. 18 indicates that the fracture morphology of the unfatigued and fatigued PET samples was similar. The sample was fatigued at a stress ratio of $R = 1/3$ with a corresponding residual strain equivalent to 2.4%. The PET sample tested without prior fatigue displayed a fracture pattern similar to fracture of fibers seen in the literature [22], a fracture initiation point followed by an opening of a v-notch, then progression to unstable fracture orthogonal to the fiber axis. The unfatigued unreinforced PET samples (Fig. 18(a)) showed a similar morphology and fracture behavior as the fatigued samples (Fig. 18(b)) (stable v-notch opening then progression to unstable fracture). However, there was a slight difference. From the fatigue process, several microfailure splits were created along the fiber axis, indicating a slight accumulation of damage in the material.

In contrast to the PET unreinforced samples, the PET-VGCNF fatigued samples exhibited a distinct fracture morphology from the unfatigued samples. Fig. 19(a) displays an unfatigued PET-VGCNF sample with no prior fatigue, demonstrating that the sample followed a standard fracture pattern. However, Fig. 19(b) and (c) display images of PET-VGCNF samples that have been fatigued at stress ratios of $R = 1/3$ with corresponding residual strains of $\varepsilon_R = 0.78\%$ and $\varepsilon_R = 0.93\%$, respectively. Defibrillation and decohesion of the reinforcing agent from the PET matrix is clearly seen in Fig. 19(b) and (c). Further, the results from Section 3 which indicate that the nanocomposite fibers experienced greater degradation than their unreinforced counterparts are supported by Fig. 19(b) and (c). In comparison with the PET fatigued sample at a residual strain of 2.4% (Fig. 18(b)), the nanocomposite fibers in Fig. 19(b) and (c) exhibited lower residual strain values of 0.78% and 0.93%, respectively. This indicates that the fatigue process in PET-VGCNF samples precipitated more fracture zones for lower residual strain values than their unreinforced counterparts, leading to lower relative residual strengths. In the next section, a discussion will be provided that seeks to further explain these changes in mechanical properties.

4. Further discussion

All of the results in Figs. 7–17 on mechanical property changes as a function of residual strain suggest that further investigation should be made into effects of maximum fatigue load levels with respect to plastic deformation. For this study, the maximum fatigue loads were similar with respect to the maximum load (stress) of the sample in uniaxial tension without prior fatigue. As stated earlier, for both PET-VGCNF and PET control samples, maximum fatigue load levels corresponded to approximately 60% of the maximum stress of the material in uniaxial tension without prior fatigue ($\sigma_{max} = 0.58\sigma_f$ for PET-VGCNF and $\sigma_{max} = 0.57\sigma_f$ for PET control). However, from a comparison of Tables 1 and 2, the PET control sample was more ductile than the PET-VGCNF sample with 5 wt.% nanofiller. This was indicated by comparing the fracture strains (ε_f) in Tables 1 and 2. The results indicate that the fracture strain for the unfatigued PET control sample was approximately 0.46 based on the samples that fractured and the SD band in Table 2. The fracture strain for the unfatigued PET-VGCNF sample was 0.28 ± 0.050. Thus in terms of testing parameters, the samples were subjected to similar maximum loads under fatigue conditions

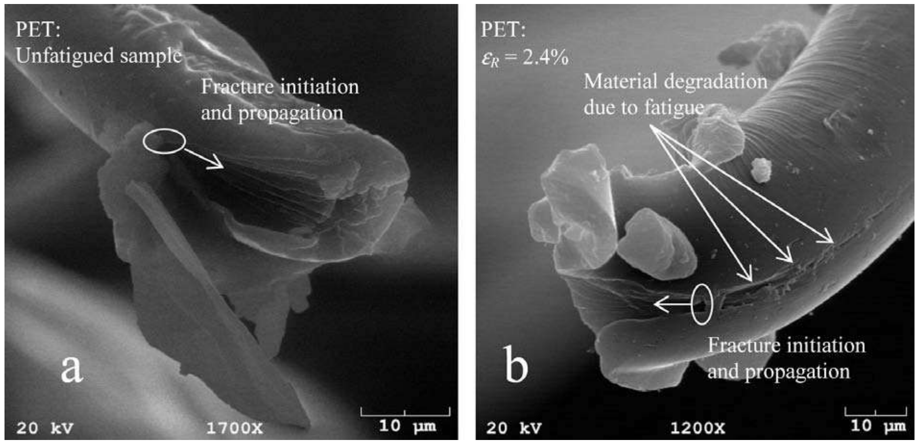

Fig. 18. (a) SEM fractograph of a PET unreinforced specimen without prior fatigue (uniaxial tensile fracture) and (b) specimen with prior fatigue and 0.024 (2.4%) residual strain from fatigue process indicated slight material degradation due to tiny microfailures along fiber axis.

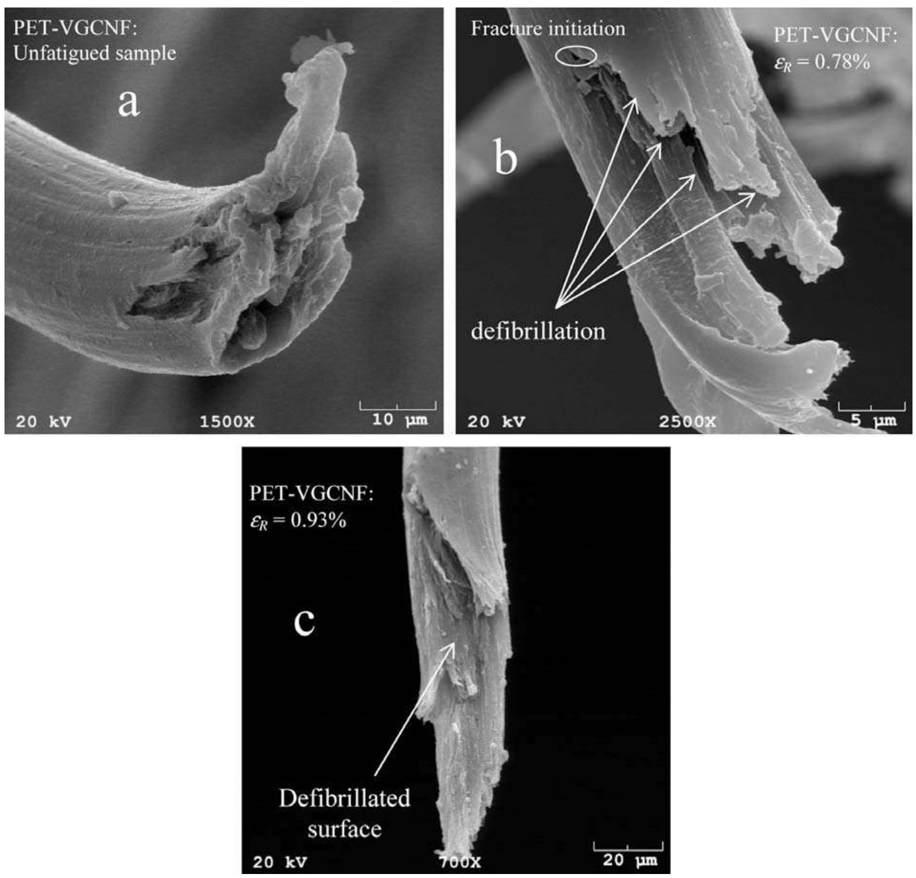

Fig. 19. (a) SEM fractograph of a PET-VGCNF specimen without prior fatigue (uniaxial tensile fracture), (b) specimen with prior fatigue and 0.0078 (0.78%) residual strain from fatigue process indicated highly distorted and tortuous crack pattern and (c) PET-VGCNF specimen with prior fatigue and 0.0093 (0.93%) residual strain from fatigue process indicating defibrillation along the fiber axis.

with respect to their fracture stress values; however, with respect to yield conditions, the samples may have experienced different types of stresses during fatigue and retained different types of residual strains. The results in Figs. 7–17 indicate that the PET samples retained both creep and plastic strains from fatigue (viscoplastic), whereas the PET-VGCNF samples may have only retained mostly creep (viscous) strains (though localized yielded regions could exist). This can be further analyzed by considering the maximum stress during fatigue for both samples in comparison to the calculated yield stress values. The PET control sample was cycled at a maximum stress equivalent to $1.5\sigma_0$, whereas the PET-VGCNF sample was cycled at a maximum stress equivalent to approximately $0.98\sigma_0$. The results indicate that cycling at stress values for similar ratios of the maximum stress for unreinforced materials and composites may engender different results for material property changes, due to the differences in the onset to plastic deformation in the samples. In fact, the literature suggests that microstructural changes occur as the result of plastic deformation, which could partially explain the difference in material property changes as a function of residual strain for the PET control and PET-VGCNF samples. Achibat et al. [23] have performed low-frequency Raman scattering measurements on shear yielded PMMA samples below the glass transition temperature (T_g) to conclude that microstructure is affected even in the case where load–unload specimens are unloaded to macroscopically identical zero stress conditions. The specimens were yielded under various conditions and the results showed that (1) anisotropy decreased in PMMA specimens cyclically loaded above the yield point with respect to undeformed specimens and (2) low-frequency Raman (LFR) data showed an excess scattered intensity in the 30–50 cm^{-1} range for plastically deformed specimens under simple shear conditions. In addition, Averett et al. [18] have shown that the yield stress subsequent to fatigue loading is clearly a function of the ratchet (accumulated) strain induced under load-controlled fatigue conditions for nylon 66 single filaments. Also, the plastic and elastic energies of filaments tested subsequent to fatigue were shown to be decreasing functions of accumulated strain. Conformational changes have also been confirmed to occur in polymers as a result of mechanical loading in the yield threshold range. Aoyama et al. [24] have observed conformational shifts from gauche to trans as a function of strain for neat PBT and PBT/rubber blends utilizing Raman spectroscopy measurements. The neat (unreinforced) PBT specimens displayed a rapid increase of gauche-to-trans conformations at approximately 10% strain, which was associated with the onset of plastic deformation (yielding) in the specimen. The reinforced samples displayed almost 100% degree of transformation from gauche to trans at a maximum strain of 50% while the unreinforced samples underwent diminutive changes of 20% degree of transformation at 50% strain. These results are similar to increases in trans content as a function of strain for poly(ethylene terephthalate) sheets that were observed from infrared (i.r.) spectroscopy studies by Cunningham et al. [25] and Hutchinson et al. [26].

From the standpoint of experimental conditions, the current research is interesting from the standpoint that changes have been observed in the mechanical properties subsequent to fatigue conducted under ambient temperature conditions. Strain induced crystallization is expected to occur in PET under mechanical loading at temperatures above T_g. Poitou et al. [27] have developed a general framework for biaxial loading of PET above T_g and linked the kinetics of crystallization with the mechanical history. Thus, one would not expect significant conformational changes in the polymer chains as a result of alternating stresses well below T_g, since both the amorphous and crystalline regions of the polymer are practically immobile.

With the aforementioned observations in mind, the following relationship is presented for residual strains in PET and PET-VGCNF samples as result of fatigue at similar maximum load levels:

For PET control samples,

$$\varepsilon_{R,cont} = \frac{\delta_R}{L_0} = \frac{\delta_c + \delta_{pl}}{L_0} \tag{4.1}$$

and for PET-VGCNF samples,

$$\varepsilon_{R,VG} = \frac{\delta_R}{L_0} = \frac{\delta_c}{L_0} \tag{4.2}$$

In (4.1) and (4.2), $\varepsilon_{R,cont}$ and $\varepsilon_{R,VG}$ represent the residual strain for the PET control and PET-VGCNF sample, respectively, δ_c represents the elongation that remained in the sample as a result of creep (viscous) deformation during constant-loading fatigue, and δ_{pl} represents the plastic elongation that was engendered in the PET control sample. For a further comparison of the results in Figs. 7–17, an interesting study would be for one to decouple the creep and plastic strains as denoted in (4.1) for the PET sample, and investigate the change in material properties as a function of residual creep strains. What is clear from this study is that PET-VGCNF fibers loaded under viscoelastic fatigue conditions suffer defibrillation and decohesion for low residual creep strains (0.78%), while the PET unreinforced samples show a similar fracture morphology even for creep strains as high as 2.4%.

5. Conclusions

Mechanical property characterization and fatigue tests have been performed on PET control and PET-VGCNF samples. In most cases, for the unfatigued samples the PET nanocomposite (PET-VGCNF) sample exhibited superior mechanical properties. To determine the influence of cyclic loading on material property changes, fatigue loading at ratios $R = 0$ and $R = 0.333$ was conducted on the PET-VGCNF and PET control samples at ratios of approximately 60% of the fracture stress. Subsequent to fatigue loading, uniaxial tensile tests were conducted on the deformed samples. The residual material properties of the PET control sample and PET-VGCNF samples were shown to be closely correlated with the residual fatigue strains. In addition, a tendency for necking was indicated by a drop in stress values for fatigued PET control and PET-VGCNF specimens, based on results from the literature.

The PET-VGCNF samples showed greater deterioration in mechanical properties as a result of fatigue compared to PET. This is supported through analysis of mechanical response and SEM images, in which defibrillation and decohesion are noticed for the fatigued PET-VGCNF samples. In fact, the PET control samples showed an increase in maximum stress, elastic modulus, hardening modulus, and tensile energy for small residual fatigue strains, supporting the claim in the literature that strain hardening occurs in early stages of cyclic extension. In addition, for both PET control and PET-VGCNF samples, the onset of yield under uniaxial tension was modified as a result of the fatigue process, as indicated by the increase in yield strain as a function residual strain. The difference in ductility and maximum fatigue stress in relation to the yield stress of the PET control and PET-VGCNF samples may indicate that the difference in mechanical property changes as a function of residual strain can be explained by the fact that the PET control sample was more susceptible to yield. When subjected to maximum fatigue loads corresponding to 60% of the fracture stress, both creep and plastic strains were engendered in the PET control samples, whereas only creep strains were engendered in the PET nanocomposite (PET-VGCNF) sample. In sum, the results from this study indicate that PET unreinforced samples can withstand a larger accumulation of strain from the fatigue process conducted at the same maximum stress level as compared with the PET-VGCNF nanocomposite samples.

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